The empirical recurrence for OEIS A302169, recovered from its tail

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Abstract

OEIS A302169 records "Empirical recurrence of order 80" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 385$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 10$, so the recurrence is proved for every $n \geq 90$. Its last coefficient is $c_{80} = 4 \neq 0$, so no further tail satisfies anything shorter; any order-80 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A302169 is "Number of 7Xn 0..1 arrays with every element equal to 0, 1 or 3 horizontally, diagonally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$64, 86, 111, 361, 621, 1563, 2853, 7306, \ \ldots$$

The entry states:

> Empirical recurrence of order 80 (see link above)

As of the "Last modified" line on the live entry (Apr 02 2018 14:53:55, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \ = \ \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 385$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 385$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 80: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 10$ is the first whose minimal order is exactly the stated 80.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 770$ exact terms from $n = 10$ on, it returns order 80 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{80} c_i\, a(n-i),$$

c_1	1	c_{21}	23995	c_{41}	16475	c_{61}	-11315
c_2	7	c_{22}	-131432	c_{42}	367508	c_{62}	-49374
c_3	-6	c_{23}	67309	c_{43}	-11576	c_{63}	-8115
c_4	15	c_{24}	226717	c_{44}	-321323	c_{64}	20729
c_5	-16	c_{25}	-62244	c_{45}	-46766	c_{65}	5377
c_6	-187	c_{26}	242094	c_{46}	-331474	c_{66}	11284
c_7	140	c_{27}	-105840	c_{47}	-18300	c_{67}	1689
c_8	25	c_{28}	-449746	c_{48}	113942	c_{68}	-6137
c_9	63	c_{29}	95149	c_{49}	46896	c_{69}	-1552
c_{10}	2069	c_{30}	-342360	c_{50}	311774	c_{70}	-1487
c_{11}	-1415	c_{31}	118486	c_{51}	27403	c_{71}	-41
c_{12}	-1816	c_{32}	636520	c_{52}	-38886	c_{72}	1054
c_{13}	416	c_{33}	-83034	c_{53}	-30210	c_{73}	208
c_{14}	-12829	c_{34}	406857	c_{54}	-246725	c_{74}	79
c_{15}	8058	c_{35}	-102867	c_{55}	-27433	c_{75}	-38
c_{16}	16232	c_{36}	-683600	c_{56}	44191	c_{76}	-82
c_{17}	-5215	c_{37}	34761	c_{57}	17554	c_{77}	0
c_{18}	50203	c_{38}	-413056	c_{58}	136751	c_{78}	-6
c_{19}	-28725	c_{39}	60995	c_{59}	19736	c_{79}	0
c_{20}	-76605	c_{40}	560120	c_{60}	-41062	c_{80}	4

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 90$, and it is the recurrence OEIS A302169 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 10$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{80} - c_1 x^{80-1} - \cdots - c_{80}$. Its constant term is $-c_{80} = -4$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 80 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 80, so $\deg q = 80$, and it is monic. It holds from some index t on. From $\max(t, 10)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 80, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 28 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 80, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 4.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-80 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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